

Load the data from runs 1-4 in MATLAB. The trigger (channel 18) marks second 2 of each trial. Information about the cue direction in each run and trial is in .mat files on Courseworks in the Experiment Files/Lab 9/Homework folder. Using this information, separate the trials into two groups: left (row 1) and right (row 2). There should be 80 left and 80 right. (1 pt)

```
%get rid of the existing data that we have.

clear; close all; clc;

run1_file = 'MotorImageryCSP_Data_Run1b_18_11_2025_11_54_09_0000.mat';
run3_file = 'MotorImageryCSP_Data_Run3_18_11_2025_12_29_07_0000.mat';
run4_file = 'MotorImageryCSP_Data_Run4_18_11_2025_12_38_31_0000.mat';

%we failed to set the file size to 100MB for run2, meaning that we have to
%concatenate them here
run2_pattern = 'MotorImageryCSP_Data_Run2_18_11_2025_12_18_09_*.mat';

% and then these are all of the files that contain the left v right
% information
cue_files = {
    'Homework/classrun1.mat';
    'Homework/classrun2.mat';
    'Homework/classrun3.mat';
    'Homework/classrun4.mat';
};

%load the 1st run
S1 = load(run1_file);
Y1 = S1.y;
fs1 = S1.SR;
trig1 = Y1(18,:);
t1 = (0:length(trig1)-1)/fs1;

%2nd run with the concatenation
files2 = dir(run2_pattern);

Y2_all = [];
trigger2_all = [];

for k = 1:length(files2)
    S2 = load(files2(k).name);
    if k == 1
        fs2 = S2.SR;
    end
    Y2_all = [Y2_all S2.y];
    trigger2_all = [trigger2_all S2.y(18,:)];
end
```

```

Y2 = Y2_all;
trig2 = trigger2_all;
t2 = (0:length(trig2)-1)/fs2;

%run3
S3 = load(run3_file);
Y3 = S3.y;
fs3 = S3.SR;
trig3 = Y3(18,:);
t3 = (0:length(trig3)-1)/fs3;

%run4
S4 = load(run4_file);
Y4 = S4.y;
fs4 = S4.SR;
trig4 = Y4(18,:);
t4 = (0:length(trig4)-1)/fs4;

%now lets make sure to pack everything into arrays for looping.
all_Y = {Y1, Y2, Y3, Y4};
all_fs = {fs1, fs2, fs3, fs4};
all_trig = {trig1, trig2, trig3, trig4};
all_t = {t1, t2, t3, t4};

%these are going to be the output structures that we have.
all_left = {};
all_right = {};
left_count = 0;
right_count = 0;

%process all 4 of the runs
for R = 1:4

    Y = all_Y{R};
    fs = all_fs{R};
    trig = all_trig{R};

    % took me a while but we have to
    % find the trigger from the rising edges (with the arrow onset = second
2)
    d = diff(trig);
    rising_idx = find(d > 0) + 1;
    rising_times = rising_idx / fs;

    %so that means that the trial onset bascially equals the trigger time -
    %2s
    trial_starts = rising_times - 2;

    % then we want to make sure that we are loading the correct cue
    variable name (z1, z2, z3, or z4)

```

```

C = load(cue_files{R});
varname = sprintf('z%d', R); %"z1", "z2", "z3", or "z4"
z = C.(varname);

% assign trials for left or right
for j = 1:40

    if z(1,j) == 1
        left_count = left_count + 1;
        all_left{left_count} = struct( ...
            'run', R, ...
            'trial', j, ...
            'start_time', trial_starts(j));
    else
        right_count = right_count + 1;
        all_right{right_count} = struct( ...
            'run', R, ...
            'trial', j, ...
            'start_time', trial_starts(j));
    end
end
end

%then just disp the final counts we get.
disp(['LEFT trials: ' num2str(left_count)]);

```

```
LEFT trials: 80
```

```
disp(['RIGHT trials: ' num2str(right_count)]);
```

```
RIGHT trials: 80
```

Filter your data from 8-20Hz and find the average power of each electrode during left versus right imagined movements. Use the time interval when the subject is actively imagining movement (4.5s to 8s after trial onset). Can you see a clear difference between the two? (6 pts)

```

clearvars -except all_left all_right all_Y all_fs all_trig all_t;
%make separte cell

%% all of the electrode names we used. All 16
electrodes = {'Fz', 'FC2', 'FC1', 'FC6', 'FC5', 'Cz', 'CP1', 'CP2', 'CP5', 'CP6', ...
              'Pz', 'P4', 'P3', 'P08', 'P07', 'Oz'};

electrode_idx = 2:17;
fs_main = 256;

%% Using the butterworth 8-20hz bandpass filter
[b,a] = butter(4, [8 20] / (fs_main/2), 'bandpass');

```

```

% then just make this a helper so that we can compute power for the one
trial
compute_power_for_trial = @(Y, fs, start_time) ...
    mean( ...
        ( ...
            filtfilt(b, a, ...
                double( ...
                    Y(electrode_idx, ...
                        round((start_time + 4.5) * fs) : ...
                        round((start_time + 8.0) * fs) ...
                    )' ...
                ) ...
            )' ...
        ).^2 ...
    , 2)';

%this is going to be for the left trials

power_left = zeros(80, 16);

for n = 1:80
    R = all_left{n}.run;
    ts = all_left{n}.start_time;
    fs = all_fs{R};
    Y = all_Y{R};

    power_left(n,:) = compute_power_for_trial(Y, fs, ts);
end

power_left_avg = mean(power_left, 1);

%and the right

power_right = zeros(80, 16);

for n = 1:80
    R = all_right{n}.run;
    ts = all_right{n}.start_time;
    fs = all_fs{R};
    Y = all_Y{R};

    power_right(n,:) = compute_power_for_trial(Y, fs, ts);
end

power_right_avg = mean(power_right, 1);

%then this will find the difference between left - right.

```

```
power_diff = power_left_avg - power_right_avg;
```

```
%plots to visualize.
```

```
%plot left trial power
```

```
figure('Color','w');
```

```
bar(power_left_avg);
```

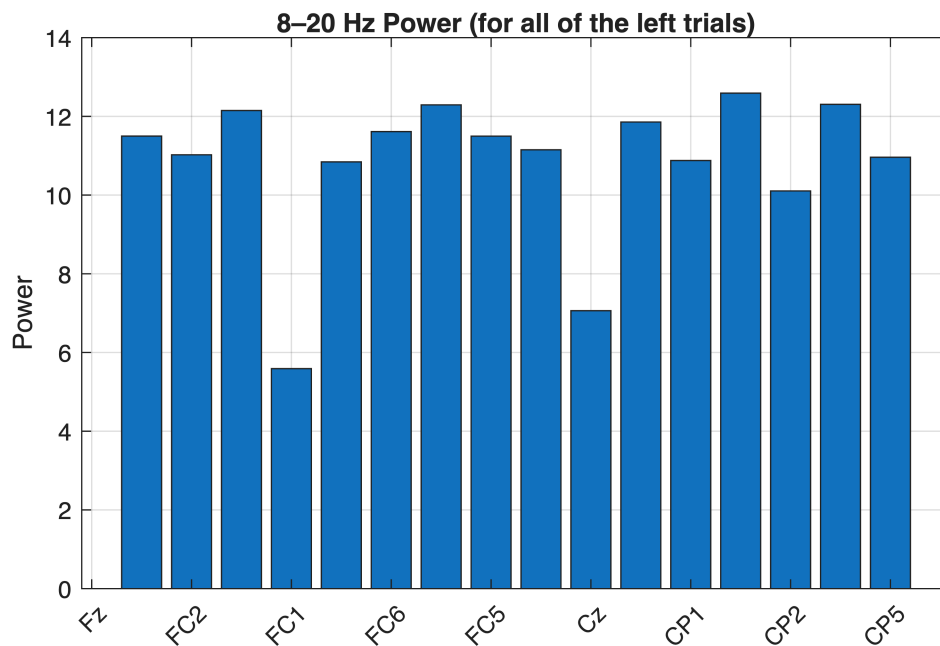
```
xticklabels(electrodes);
```

```
xtickangle(45);
```

```
ylabel('Power');
```

```
title('8-20 Hz Power (for all of the left trials)');
```

```
grid on;
```



```
%Right trial power
```

```
figure('Color','w');
```

```
bar(power_right_avg);
```

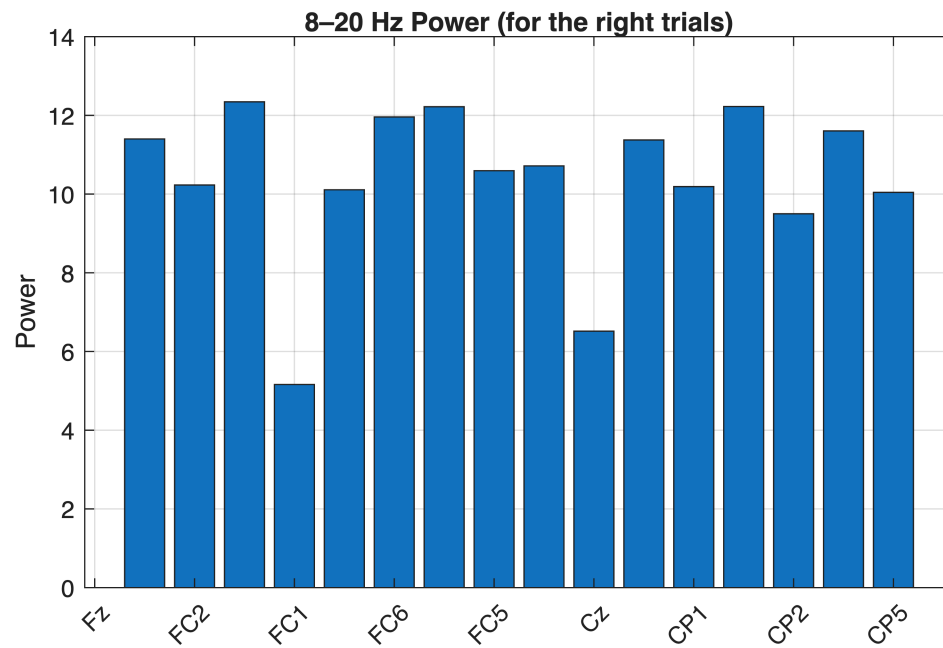
```
xticklabels(electrodes);
```

```
xtickangle(45);
```

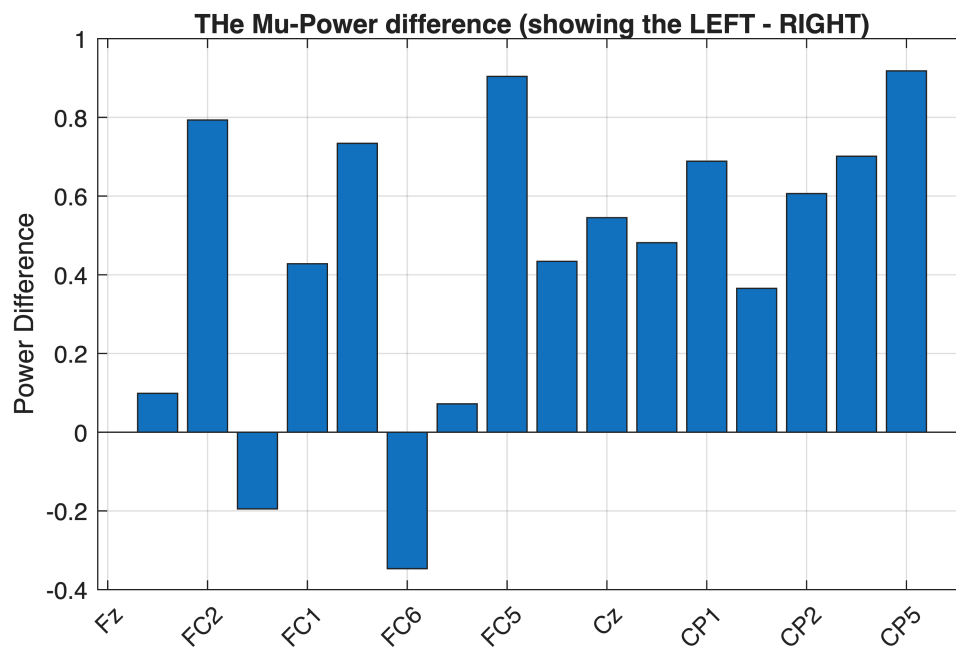
```
ylabel('Power');
```

```
title('8-20 Hz Power (for the right trials)');
```

```
grid on;
```



```
%Left - Right
figure('Color','w');
bar(power_diff);
xticklabels(electrodes);
xtickangle(45);
ylabel('Power Difference');
title('The Mu-Power difference (showing the LEFT - RIGHT)');
grid on;
```



```
%sanity / debug
disp('Average Power LEFT');
```

Average Power LEFT

```
disp(power_left_avg);
```

11.4986 11.0234 12.1482 5.5899 10.8430 11.6122 12.2915 11.4975 11.1503 7.0617 11.4986

```
disp('Average Power RIGHT');
```

Average Power RIGHT

```
disp(power_right_avg);
```

11.3999 10.2302 12.3431 5.1618 10.1090 11.9589 12.2195 10.5936 10.7161 6.5166 11.3999

```
disp('Difference (L - R)');
```

Difference (L - R)

```
disp(power_diff);
```

0.0987 0.7932 -0.1949 0.4281 0.7340 -0.3468 0.0720 0.9039 0.4342 0.5451 0.0987

I believe that the mu-band analysis reveals what I would consider to be a noticeable difference between left and right-hand motor imagery. It appears that the right-hand trials are showing a consistent drop in the 8–20 Hz power across several of the right and hemisphere electrodes (including FC6, CP6, and P4), which would then be somewhat indicative of some sort of strong mu-rhythm suppression that was taking place during the right-hand imagination by our subject. It then follows that this would end up creating a clear and lateralized pattern that would be visible on the right side of the scalp. In contrast, we can see that the left-hand imagery does not come together to produce an equally clean effect on the left hemisphere (since we can observe that some of the electrodes, like FC5, CP5, and P3, show weaker or less stable reductions in their respective powers. In my research for this homework, it seems that this asymmetry is typical in these sorts of motor-imagery tasks, where one hand (often the dominant one, which in this case was true) generates stronger and also more reliable neural engagement. Overall, it appears that the data demonstrate a clear separability between our conditions, with the right-hand imagery seeming to produce the more robust and the more spatially coherent suppression pattern.

Calculate the difference between the two conditions for all electrodes and display the subtraction on the scalp map. You can find the new electrode configuration file 'CSP.locs' in Courseworks. In which parts of the brain do you observe a bigger difference? (3 pts)

```
%you have to run both of these, should work if you run the notebook all in
%one go
addpath('/Users/alexanderspeer/Desktop/BCI lab/eeglab14_1_2b'); % i guess
adjust if needed
eeglab; %topoplot
```

```
Warning: Adding folders named 'resources' to the path is not supported: /Users/alexanderspeer/
Desktop/BCI_lab/eeglab14_1_2b/functions/resources
eeglab: options file is ~/eeg_options.m
EEGLAB: error while adding plugin "eegplugin_dipfit"
    Error using uimenu
    First argument must be a valid parent, such as a Figure or Panel object.
EEGLAB: error while adding plugin "eegplugin_firfilt"
    Error using uimenu
    First argument must be a valid parent, such as a Figure or Panel object.
```

```
electrodes = {'Fz', 'FC2', 'FC1', 'FC6', 'FC5', 'Cz', 'CP1', 'CP2', 'CP5', 'CP6', ...
              'Pz', 'P4', 'P3', 'P08', 'P07', 'Oz'};
```

```
%load the channel locations, nothing to change since using all of the them
locs_path = 'BCI.locs'; %from the courseworks
chanlocs_full = readlocs(locs_path);
```

```
readlocs(): 'loc' format assumed from file extension
```

```
% match to them by labell
loc_labels = {chanlocs_full.labels};
[tf, selLocIdx] = ismember(electrodes, loc_labels);

if any(~tf)
    missing = electrodes(~tf);
    error('not found in BCI.locs: %s', strjoin(missing, ', '));
end

chanlocs = chanlocs_full(selLocIdx);

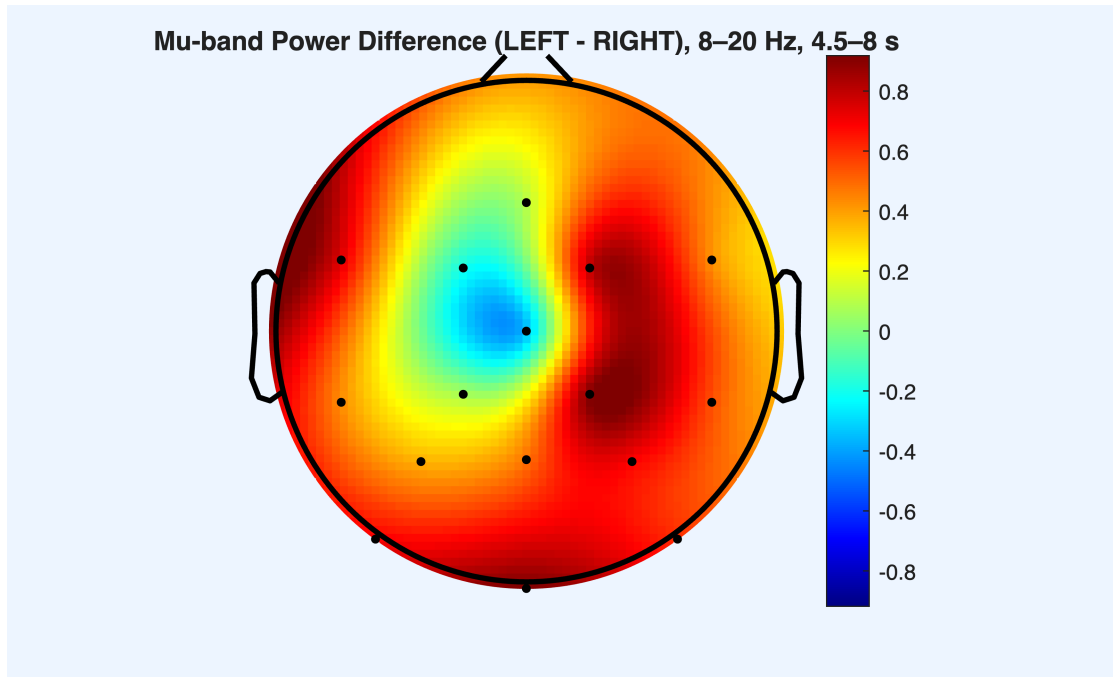
% scalp map
vals = power_diff; % same thing as before left - right, with the length 16

figure('Color','w');
topoplot(vals, chanlocs, ...
    'electrodes','on', ...
    'numcontour', 6, ...
    'style','map' ...
);
```

```
Warning: When plotting pvalues in totoplot, use option 'conv' to minimize extrapolation effects
```

```
colorbar;
title('the Mu-band Power Difference (LEFT - RIGHT), @ 8-20 Hz, and then
lasting 4.5-8 s');

%trying symmetric colors
mx = max(abs(vals));
caxis([-mx mx]);
```

It seems to me that (similar to our earlier findings) the scalp map shows that the biggest difference between imagining left and right hand movement appears over the left side of the brain, near the top of the head, which, as we discussed previously, is where the brain controls movements of the right hand. On this topoplot, this area is shown in dark blue on the map, which would then mean that the right-hand imagery ended up causing a noticeably stronger drop in the mu-band power than in the left-hand imagery.

Once again, this fits what we have come to expect biologically. When someone imagines moving their right hand, it is going to be the left sensorimotor cortex that ends up becoming more active and, as well as having its rhythmic activity decrease. Overall, it appears that the most meaningful and also the most clearly defined difference is located over the left sensorimotor region, which lines up with how the brain would normally go about organizing such hand movement control.

All in all, the results strongly suggest that the subject is just better at visualizing this right-hand movement. To this end, we can see that the right-hand imagery managed to produce clearer and stronger suppression of mu-band activity over the left sensorimotor cortex, which is exactly where we would expect the brain to activate when imagining movements of the right hand. In contrast, the responses to left-hand imagery were weaker and less focused. I would say that this asymmetry makes even more sense given that the subject is both right-handed and also plays violin at Juilliard, meaning that their right hand is likely far more trained, more coordinated, and overly represented more dominantly in the brain.